MIL-T-60102A(MU)
18 July 1974
SUPERSEDING
MIL-T-60102(MU)
24 January 1964

MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M9 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M9, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings

DRAWINGS

Armament Command

F10520479	- Thruster, Cartridge Actuated, M9 Metal Parts Assembly
C8845196	- Box, Fiberboard (Inner) Packing Ammunition for Thruster, Cartridge Actuated, M9

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

C8845197	- Box, Packing Ammunition for Thruster, Cartridge Actuated M9
C9217287	- Box, Fiberboard (Outer) Alternative Packing, Ammunition for Thruster, Cartridge Actuated M9
IEL-10520479	- Inspection Equipment List for Thruster, Cartridge Actuated M9, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific government functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Mechanical properties.-Mechanical properties required by the drawings shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.3 Hydrostatic pressure.-The body shall withstand a hydrostatic pressure of 10,000 pounds per square inch (PSI), minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

3.4 Metal parts assembly.-Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following:

3.4.1 All threaded joints shall be assembled hand tight, except as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

3.4.2 Torque.-Torque requirements shall be as specified in Table I. No movement or deformation shall occur upon application of the breakaway torque.

TABLE I

Components	A Seating or Bottoming (inch pounds)	B Breakaway (inch pounds)
Piston and Adapter	100 ± 20	75 min.

3.4.3 Trunnion rotation.-Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.4.4 Piston rotation.-Upon completion of assembly, the adapter shall be free to rotate 360 degrees in a clockwise direction.

3.5 Protective finish.-Protective finish shall be in accordance with the applicable specifications and drawings, and shall apply to all surfaces of the component regardless of contour configuration.

3.6 Residual magnetism.-The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.7 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.-All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.7.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.-No part or assembly shall contain chips, dirt, grease, rust, corrosion or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Governmant. The Government reserves the right to perform any of the inspections set forth in the specification and drawing where such inspections are deemed necessary to assure supplied and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550 and MIL-I-45208, except as specified herein.

4.2 First artical sample.-Prior to initiation of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 5 metal part assemblies, 10 sets of unassembled parts and 50 shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1 Examination and tests.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.3.1.2 Size of lot.-The shipping lots of the metal parts assemblies shall consist of 1200 thrusters maximum provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed in ANSI-B46.1-N2, shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:19Z
Check the source to verify that this is the currentversion before use.

TABLE II

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 to 300	100% Examination	-	-	-	-
301 to 1300	110	1	2	2	3
1301 to 3200	150	2	3	3	4
3201 to 8000	225	3	4	4	5
8001 to 22000	300	4	5	5	6

TABLE III

	Total allowable defects per sample size			
Sample Size for Major Defects - - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects - - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Adapter, Drawing C10520480, covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of external thread, min.	Gage
102. Major diameter of external thread, min.	Gage
103. Pitch diameter of internal thread, max.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.1 (Cont'd)

	Defect	Method of Inspection
104.	Minor diameter of internal thread, max.	Gage
105.	Concentricity of pitch diameter of external thread with pitch daimeter of internal thread	Gage
106.	Concentricity of boss with pitch diameter of internal thread	Gage
107.	Thread damaged or not full	Visual
108.	Metal defective	Visual
109.	Protective coating inadequate or defective	Visual
110.	Burr	Visual
111.	Foreign matter	Visual

MINOR

201.	Pitch diameter of internal thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Diameter of undercut, internal thread	Gage
204.	Diameter of undercut, external thread	Gage
205.	Diameter of boss	Gage
206.	Depth of thread cavity	Gage
207.	Total length	Gage
208.	Length of flange	Gage
209.	Height of boss	Gage
210.	Diameter of flange	Measure
211.	Width across flats	Measure
212.	Width of undercut, external	Measure
213.	Width of undercut, internal	Measure
214.	Location of flats	Measure
215.	Marking missing, incorrect or illegible	Visual
216.	Radius or chamfer missing or incorrect	Visual
217.	Surface finish, improper	Visual

4.3.2.2 Body, Drawing F10520482, covering a detail of Drawing F10520479.

CRITICAL

None defined

MAJOR

101.	Pitch diameter of large thread, max.	Gage
102.	Minor diameter of large thread, max.	Gage
103.	Pitch diameter of thread, bypass pert, max.	Gage
104.	Minor diameter of thread, bypass port, max.	Gage

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:19Z
Check the source to verify that this is the currentversion before use.

4.3.2.2 (Cont'd)

	Defect	Method of Inspection
105.	Concentricity of thread cavity with long cavity	Gage
106.	Concentricity of thread cavity with large thread	Gage
107.	Concentricity of head with large thread	Gage
108.	Concentricity of bore with long cavity	Gage
109.	Concentricity of grooves with bore	Gage
110.	Concentricity of counterbore with bore	Gage
111.	Concentricity of pin holes	Gage
112.	True position of pin holes	Gage
113.	Diameter of long cavity, max.	Gage
114.	Diameter of bore, max.	Gage
115.	Diameter of pin holes	Gage
116.	Diameter of grooves	Gage
117.	Diameter of threaded cavity, max.	Gage
118.	Diameter of internal undercut, max. if applicable	Gage
119.	Depth of cavity, bypass port	Gage
120.	Depth of counterbore, bypass port	Gage
121.	Length of thread, bypass port	Measure
122.	Diameter of chamfer, bypass port	Measure
123.	Angle of chamfer, bypass port	Measure
124.	Internal radii forward eand, long cavity	Visual
125.	Thread damaged or not full	Visual
126.	Metal defective	Visual
127.	Protective finish inadequate or defective	Visual
128.	Surface finish or long cavity and chamfer of bypass port, improper	Visual
129.	Burr or sharp corners	Visual
130.	Foreign matter	Visual

MINOR

201.	Pitch diameter of large thread, min.	Gage
202.	Pitch diameter of thread, bypass port, min.	Gage
203.	Pitch diameter of set screw thread	Gage
204.	Minor diameter of set screw thread, max.	Gage
205.	Diameter of long cavity, min.	Gage
206.	Diameter of bore, min.	Gage
207.	Diameter of thread cavity, min.	Gage
208.	Diameter of large thread undercut, max.	Gage
209.	Diameter of body	Gage
210.	Diameter of counterbore set screw end	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.2 (Cont'd)

	Defect	Method of Inspection
211.	Diameter of relief undercut, bypass port	Gage
212.	Diameter of hole in bypass port	Gage
213.	Diameter of shoulder	Gage
214.	Diameter of head	Gage
215.	Diameter of counterbore, bypass port	Gage
216.	Diameter of bypass port	Gage
217.	Depth to bottom of large cavity	Gage
218.	Depth to bottom of thread undercut	Gage
219.	Depth to bottom of large cavity	Gage
220.	Depth of counterbore	Gage
221.	Length of effective set screw thread, min.	Gage
222.	Total length	Gage
223.	Length to shoulder	Gage
224.	Length of set screw thread, max.	Gage
225.	Location of pin holes	Gage
226.	Location of groove rear long cavity	Gage
227.	Location of groove near shear pin hole	Gage
228.	Location of relief undercut	Gage
229.	Location of bypass port	Gage
230.	Height of bypass port	Measure
231.	Width of internal undercut, max., if applicable	Measure
232.	Width of groove	Measure
233.	Width of thread undercut	Measure
234.	Width of relief undercut	Measure
235.	Length to end of undercut	Measure
236.	Profile of grooves	Measure
237.	Profile of bypass port	Measure
238.	Marking, missing, incorrect or illegible	Visual
239.	Radius or chamfer missing or incorrect, except as otherwise classified	Visual

4.3.2.3 Breech, Drawing D10520483, covering a detail of Drawing F10520479.

CRITICAL

None defined

MAJOR

101.	Pitch diameter of thread on head, min.	Gage
102.	Major diameter of thread on head, min.	Gage
103.	Pitch diameter of thread on body, min.	Gage
104.	Major diameter of thread on body, min.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

	Defect	Method of Inspection
105.	Pitch diameter of internal thread, max.	Gage
106.	Minor diameter of internal thread, max.	Gage
107.	Concentricity of groove with body	Gage
108.	Concentricity of body with body thread	Gage
109.	Concentricity of head thread with body thread	Gage
110.	Concentricity of long cavity with body thread	Gage
111.	Concentricity of striker hole with long cavity	Gage
112.	Concentricity of boss cavity with long cavity	Gage
113.	Concentricity of boss with body thread	Gage
114.	True position of pin hole	Gage
115.	Perpendicularity of boss face with body thread	Gage
116.	Diameter of boss	Gage
117.	Diameter of long cavity, max.	Gage
118.	Diameter of body, min.	Gage
119.	Diameter of pin hole, max.	Gage
120.	Diameter of groove	Gage
121.	Length of internal thread	Gage
122.	Height of boss	Gage
123.	Profile of groove	Measure
124.	Thread damaged or not full	Visual
125.	Escapement hole missing	Visual
126.	Metal defective	Visual
127.	Protective finish inadequate or defective	Visual
128.	Surface finish or chamfer, threaded cavity and long cavity, improper	Visual
129.	Burr or sharp edge	Visual
130.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread on head, max.	Gage
202.	Pitch diameter of thread on body, max.	Gage
203.	Pitch diameter of internal thread, min.	Gage
204.	Diameter of long cavity, min.	Gage
205.	Diameter of body, max.	Gage
206.	Diameter of pin hole, min.	Gage
207.	Diameter of counterbore, threaded cavity	Gage
208.	Diameter of striker hole	Gage
209.	Diameter of boss cavity	Gage
210.	Diameter of thread undercut, min.	Gage
211.	Diameter of head undercut, min.	Gage
212.	Diameter of escapement hole	Gage
213.	Diameter of wrench holes	Gage
214.	Depth of threaded cavity	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

	Defect	Method of Inspection
215.	Depth of counterbore, threaded cavity	Gage
216.	Total length	Gage
217.	Location of groove	Gage
218.	Location of pin hole	Gage
219.	Location of escapement hole	Gage
220.	Location of wrench hole	Gage
221.	Thickness of through boss	Gage
222.	Thickness at bottom of boss cavity	Gage
223.	Thickness of head	Gage
224.	Depth of wrench holes	Measure
225.	Countersink of escapement holes	Measure
226.	Diameter of chamfer, threaded cavity	Measure
227.	Width of thread undercut	Measure
228.	Width of head undercut	Measure
229.	Width of groove	Measure
230.	Marking missing, incorrect or illegible	Visual
231.	Radius or chamfer missing or incorrect	Visual
232.	Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Collar, Drawing C10520484, covering a detail of Drawing F10520479.

CRITICAL

None defined

MAJOR

101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Concentricity of cavity with pitch diameter of thread	Gage
104.	Symmetry of flats with pitch diameter of thread	Gage
105.	Diameter of cavity, max.	Gage
106.	Metal defective	Visual
107.	Protective finish inadequate or defective	Visual
108.	Burr	Visual
109.	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the currentversion before use.

4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, min.	Gage
202. Diameter of cavity, min.	Gage
203. Diameter, outside	Gage
204. Depth of cavity	Gage
205. Total length	Gage
206. Width across flats	Gage
207. Length of flats	Measure
208. Marking missing, incorrect or illegible	Visual
209. Radius or chamfer missing or incorrect	Visual
210. Surface finish, improper	Visual

4.3.2.5 Pin, Firing, Drawing C10520485, covering a detail of Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of striker with body	Gage
102. Concentricity of groove with body	Gage
103. Concentricity of striker shoulder with body	Gage
104. True position of pin hole	Gage
105. Diameter of body, min.	Gage
106. Diameter of striker	Gage
107. Diameter of groove	Gage
108. Diameter of pin hole, max.	Gage
109. Profile of striker	Measure
110. Profile of groove	Measure
111. Metal defective	Visual
112. Protective finish inadequate or defective	Visual
113. Surface finish of striker, improper	Visual
114. Burr or sharp edges	Visual
115. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of pin hole, min.	Gage
203. Diameter of striker shoulder	Gage
204. Length of striker	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.5 (Cont'd)

Defect	Method of Inspection
205. Length from body to tip of striker	Gage
206. Total length	Gage
207. Location of pin hole	Gage
208. Location of groove	Gage
209. Depth of pin hole	Measure
210. Depth of slot in base	Measure
211. Width of groove	Measure
212. Width of slot in base	Measure
213. Location of slot in base	Measure
214. Marking missing, incorrect or illegible	Visual
215. Radius or chamfer, missing or incorrect	Visual
216. Surface finish, improper, except as otherwise classified	Visual

4.3.2.6 Pin, Shear, Firing Pin, Drawing B10520486, covering a detail of Drawing F10520479.

CRITICAL

None defined

MAJOR

101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corner	Visual
106. Foreign matter	Visual

MINOR

201. Total length, max.	Gage

4.3.2.7 Pin, Shear, Piston, Drawing B10520487, covering a detail of Drawing F10520479.

CRITICAL

None defined

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:19Z
Check the source to verify that this is the currentversion before use.

4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr or sharp corner	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.3.2.8 Piston, Drawing D10520488, covering a detail of Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of shoulder with body	Gage
104. Concentricity of head with body	Gage
105. Concentricity of groove with head	Gage
106. Concentricity of pitch diameter of thread with body	Gage
107. Diameter of head	Gage
108. Diameter of groove	Gage
109. Diameter of body, min.	Gage
110. Diameter of shoulder, min.	Gage
111. Profile of groove	Measure
112. Thread damaged or not full	Visual
113. Metal defective	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish, body and shoulder, improper	Visual
116. Burr or sharp edge	Visual
117. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of body, max.	Gage
203. Diameter of shoulder, max.	Gage
204. Diameter of cavity	Gage
205. Diameter of escapement hole in head	Gage
206. Length of shoulder	Gage
207. Length from head to thread undercut	Gage
208. Total length	Gage
209. Location of groove	Gage
210. Location of holes thru body	Gage
211. Angular location of escapement hole in head	Measure
212. Countersink of holes thru body	Measure
213. Diameter of thread undercut	Measure
214. Diameter of undercut at head, min. , if applicable	Measure
215. Diameter of holes thru body	Measure
216. Depth of cavity	Measure
217. Width of groove	Measure
218. Width of head undercut, max., if applicable	Measure
219. Width of thread undercut	Measure
220. Marking missing, incorrect or illegible	Visual
221. Radius or chamfer, missing or incorrect	Visual
222. Surface finish improper, except as otherwise classified	Visual

4.3.2.9 Ring, Drawing C10520489, covering a detail of Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of inside diameter with outside diameter	Gage
102. Concentricity of pin holes	Gage
103. True position of pin hole	Gage
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Burr	Visual
107. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.9 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Outside diameter	Gage
202. Inside diameter	Gage
203. Diameter of pin holes	Gage
204. Depth of pin holes	Gage
205. Width	Gage
206. Location of pin holes	Gage
207. Marking missing, incorrect or illegible	Visual
208. Radius or chamfer, missing or incorrect	Visual
209. Surface finish, improper	Visual

4.3.2.10 Trunnion, Drawing D10520490, covering a detail of Drawing F10520479.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of lugs	Gage
102. True position of lug	Gage
103. Symmetry of flats at root of lugs with inside diameter	Gage
104. Symmetry of faces with inside diameter	Gage
105. Diameter of lugs, min.	Gage
106. Diameter of inside, max.	Gage
107. Width across flats, min.	Gage
108. Metal defective	Visual
109. Protective finish inadequate or defective	Visual
MINOR	
201. Diameter of lugs, max.	Gage
202. Diameter of inside, min.	Gage
203. Length of lugs	Gage
204. Width across flats, max.	Gage
205. Location of lug	Gage
206. Thickness of body	Gage
207. Thickness of wall	Gage
208. Marking missing, incorrect or illegible	Visual
209. Radius or chamfer, missing or incorrect	Visual
210. Surface finish improper	Visual
211. Burr or sharp corner	Visual
212. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.11 Adapter and Pellet Assembly, Drawing C10520481, covering a detail of Drawing F10520479.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Protrusion of pellets over thread, min.	Gage
102. Pellet loose, missing or improperly inserted	Visual
103. Thread damaged or not full	Visual
104. Foreign matter	Visual
MINOR	
201. Location of pellets	Measure

4.3.2.12 Breech and Firing Pin, Sub-Assembly, consisting of Breech 10520483, Firing Pin 10520485 and "O" Ring MS-28775-009, covering a detail of Dwg. F10520479.

CRITICAL	
1. Firing pin protrusion	Gage
MAJOR	
101. Thread damaged or not full	Visual
102. Protective finish, inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.3.2.13 Assembly, Drawing F10520479.

CRITICAL

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.13 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Trunnion not free to rotate 360 degrees	Manual
102. Piston not free to rotate 360 degrees clockwise	Manual
103. Any part missing, except as otherwise classified	Visual
104. Improper assembly	Visual
105. Thread damaged or not full	Visual
106. Foreign matter	Visual
MINOR	
201. Thread compound or set screw missing	Visual
202. Set screws not filled with barrier material	Visual
203. Shipping plugs or shipping cap	Visual
204. Nylok pellet missing	Visual
205. Protective finish, inadequate or defective	Visual
206. Burr	Visual

4.3.2.14 Box, Fiberboard, Inner, Drawing C8845196.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Box broken or damaged	Manual
203. Number of units contained incorrect	Manual

4.3.2.15 Box, Packing, Drawing C8845197.

CRITICAL	
None defined	
MAJOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.2.15 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Strapping missing, loose, broken, or improperly assembled	Manual
202. Contents loose	Manual
203. Box damaged	Visual
204. Contents exposed	Visual
205. Marking misleading or unidentifiable	Visual
206. Number of units incorrect	Visual

4.3.2.16 Carton Alternative, Packing, Drawing C9217287.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Carton broken or damaged	Manual
203. Number of units incorrect	Visual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4, except as provided in Specification MIL-I-45208.

4.3.3.1 Hardness.-Fifty units from each heat treated lot of components shall be tested for compliance with the values noted on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Hydrostatic pressure.-Each body shall be subjected to the internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.5 shall be cause for rejection of the component.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.3.3.3 Shear pin strength.-Fifty pins of each type from each lot shall be tested for shear strength. Failure of any shear pin to comply with the requirements of the drawing, as applicable, shall be cause for rejection of the lot.

4.3.3.4 Breakaway torque and curing time.-Each piston and adapter shall be subjected to a breakaway torque application after the thread sealant has been cured. Failure of any unit to comply with the requirements of 3.6.2, Table I and Drawing F10520479, as applicable, shall be cause for rejection of the unit.

4.3.3.5 Protective finish.-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable specification listed on the drawing.

4.3.3.6 Residual magnetism.-Each assembly shall be subjected to a test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.8 shall be cause for the rejection of the assembly.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-10520479, except as provided in Specification MIL-I-45208.

4.4 Test methods and procedures.

4.4.1 Hardness.-Prior to the application of protective finish, a hardness testing machine shall be used to determine the hardness values specified on the detail drawing.

4.4.2 Hydrostatic pressure.-The test shall be performed prior to the application of protective finish. The testing device used shall internally seal the body at each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength.-After application of protective finish, the load shall be applied at the steady rate of 0.125 inches per minute.

4.4.4 Breakaway torque and curing time.-The piston shall be secured in an air vise or other holding fixture. The torque shall be applied to the adapter. Curing shall be at room temperature for the length of times as specified on Drawing F10520479.

4.4.5 Protective finish.-Test procedure for protective finish shall be in accordance with the specification referenced on the component drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

4.4.6 Residual magnetism.-The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the thruster 6 inches from and at the same height as the compass in a north-south horizontal position with the compass set in an east-west heading as shown in Figure I. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity. Level A - Drawings C8845196 and C8845197.
Level B - Drawings C8845196 and C9217287.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1377-A585

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-60102A(MU), Thruster, Cartridge Actuated, M9 Metal Parts Assembly

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:19Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Philadelphia, PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:19Z
Check the source to verify that this is the current version before use.